

ACTIVITY 8

HackerRank Algorithmic Problem-Solving & Portfolio Integration

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Program	B.Tech – CSIT
Semester	3rd Semester

HACKERRANK PROFILE

<https://www.hackerrank.com/profile/shriharidesai209>

GITHUB REPOSITORY

<https://github.com/shriharidesai209-commits/HackerRank-3rdSem-Portfolio>

PROBLEMS COMPLETED

No.	Problem	Time	Space
1	Diagonal Difference	$O(N)$	$O(1)$
2	Dynamic Array	$O(N + Q)$	$O(N)$
3	Time Conversion	$O(1)$	$O(1)$
4	Compare the Triplets	$O(1)$	$O(1)$
5	Sparse Arrays	$O(N + Q)$	$O(N)$

HACKERRANK ACHIEVEMENTS

Achievement	Status
Problem Solving	3-Star
C++ Programming	2-Star

All five mandatory HackerRank problems were completed and documented in the public GitHub portfolio repository.

REFLECTIVE SUMMARY

Algorithmic Problem-Solving & Optimization Techniques

This activity helped me strengthen my understanding of algorithmic problem-solving and the importance of writing efficient solutions. By solving five HackerRank problems involving arrays, matrices, strings, dynamic arrays, and frequency mapping, I practiced converting problem statements into clear computational steps. I learned that choosing an appropriate data structure can significantly improve the efficiency of a solution. For example, using dynamic arrays and hash-based frequency mapping helps process larger inputs efficiently instead of repeatedly searching through data. I also learned to analyze both time and space complexity before considering a solution complete. Comparing approaches based on $O(N)$, $O(N + Q)$, and $O(1)$ complexity helped me understand how algorithms behave as input size increases. The activity also improved my ability to handle edge cases, such as AM/PM conversion and diagonal calculations. Completing the problems on HackerRank gave me practical experience with automated test cases and verifying whether my implementation was correct. Finally, organizing the solutions, explanations, complexity analysis, and submission evidence on GitHub helped me understand how problem-solving work can be presented as part of a professional technical portfolio. Overall, this activity improved both my coding practice and my understanding of writing structured, efficient, and well-documented solutions.

PORTFOLIO LINKS

HackerRank Profile	https://www.hackerrank.com/profile/shriharidesai209
GitHub Repository	https://github.com/shriharidesai209-commits/HackerRank-3rdSem-Portfolio

Repository: HackerRank-3rdSem-Portfolio